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St. Joseph's college f Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore–607001

B. Sc DEGREE EXAMINATION-APRIL 2021

EBC 510A- PLANT BIOCHEMISTRY

Time: 3 hrs **Max Marks: 75**

SECTION-A (25X1=25)

Answer ALL the questions

I. choose the best answer: (10X1=10)

1. Plasmodesmata is usually observed between
   a. sieve tubes and bast fibres
   b. trachea and phloem fibres
   c. xylem parenchyma and xylem fibres
   d. sieve tubes and companion cells
2. The rapid type of water absorption is
   a. Active absorption
   b. Passive absorption
   c. Active osmotic absorption
   d. Active non-osmotic absorption
3. Highest auxin concentration occurs
   a. In growing tips b. In leaves
   c. At base of plant organs d. In xylem and phloem
4. What is the primary precursor of IAA?
   a. Phenylalanine b.Tyrosine
   c. Tryptophan d.Leucine

5. The radiant energy absorbed by carotenoids is
   a. Lost as heat
   b. Transferred to chlorophylls
   c. Re-emitted ass fluorescence or phosphorescence
   d. Used to perform photochemical act.

6. In C4 plants, Calvin cycle operates in
   a. Stroma of bundle sheath chloroplasts
   b. Grana of bundle sheath chloroplasts
   c. Grana of mesophyll chloroplasts
   d. Stroma of mesophyll chloroplasts

7. A simple mechanical tissue devoid of lignin is
   a. Parenchyma b. Sclerenchyma
   c. Collenchymas d. Chlorenchyma

8. Morphine, nicotine and cocaine are the examples of
   a. Alkaloids b. Terpenoids
   b. Steroids d. Tannins

9. A red pigment present in the root nodules of leguminous plants is known as
   a. Phycoerythrin b.Bacteriochlorophyll
   c. Leghaemoglobin d.Bacterioviridin

10. Component of nitrogenase and nitrate reductase is:
   a. N b.Mo
   b. Co d.No specific component

II. True or False (5X1=5)

11. Meristem cells are undifferentiated.
12. Sun light is not essential for germination.
13. In photosynthesis water is reduced.
14. Hardness of woody tissue is due to cellulose.
15. Most plants and animals cannot use the nitrogen in nitrogen gas.

III. Answer in One or Two Sentences (10X1=10)

16. What is Ion Exchange?
17. Define cell wall?
18. What are the uses of cytokinin?
19. Name the fruit ripening hormone.
20. Define chlorophyll.
21. What is CAM?
22. What is pectin?
23. Define salinity of water.
24. Name any 2 symbiotic nitrogen fixing bacteria.
25. Write about nif gene.

SECTION-B (4X5=20)

Answer any FOUR of the following:

26. Explain Gibbs–Donnan effect.
27. Write briefly about fruit ripening hormones.
28. Describe Hatch-Slack pathway.
29. List out the light absorbing pigments. Explain.
30. Explain the structure and functions of lignin.

31. Discuss about nif regenes.

SECTION-C (3X10=30)

Answer any THREE of the following:

32. Describe the structure and functions of plant cell.
33. Explain the functions of gibberellins, cytokinins and IAA.
34. Discuss in detail about C_3Pathway.
35. Explain secondary metabolites with examples.
36. Elaborately describe the symbiotic nitrogen fixing bacteria.

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